

The Social Stratification of Housing Wealth: Bringing Socio-Spatial Differentiation through the Price Gap Index

Jean-Sauveur Ay¹ and Thibault Le Corre^{1,2}

Abstract

The spatio-temporal variation of housing prices is central to study the distribution of housing wealth (HW) across social groups. This paper presents a price decomposition framework to analyze social stratification of HW in terms of housing characteristics, location, and transaction date across labor-based occupational categories (OCs) of homeowners. Beyond differentiated housing prices and capital gains, we highlight the importance of the concept of Price Gap Index (PGI) – the difference between average buyer and seller prices for each OC – to catch the redistribution of HW induced by housing transactions. Using data over two decades (1998–2017) for the French region *Bourgogne-Franche-Comté*, we find that static HW differences reflect the usual social hierarchy of the labor market. However, the PGI decomposition shows more regressive results of HW accumulation across OCs than the capital gains computed from usual housing price index over time. These results confirm that socio-spatial dynamics on the housing markets is an important inequality generator acting through redistribution of HW during transactions.

Keywords

housing wealth; socioeconomic analysis; prices; inequalities; socio-spatial dynamics

Introduction

Owing to a twofold movement of financialization and rentierization, housing prices have risen dramatically in most OECD countries over the last three decades (Renaud and Kim 2008; Fernandez and Aalbers 2017). Fueled by this inflation, capital returns from homeownership increased sharply, reinforcing housing as a major asset for households (Schwartz and Seabrooke 2008). In this context, HW becomes a central feature of inequalities (Adkins, Cooper, and Konings 2019).

Since Rex and Moore (1967), homeownership has been widely debated in social science studies, especially by scholars seeking to identify housing-related systems of social stratification (Saunders 1984; Badcock 1989; Thorns 1989). These works hold particular significance today as returns from housing capital grew faster than labor incomes (Piketty and Zucman (2014); Ryan-Collins and Murray (2021). Taking seriously the assumption of a housing asset-based stratification (Adkins, Cooper, and Konings (2019), social inequalities stemming from housing markets must be evaluated accurately and compared with more extensively studied inequalities from labor markets.

Surprisingly, evidence and methodological frameworks to better grasp inequalities in HW remain scarce (Wind and Hedman 2018; Arundel and Lennartz 2020). Typical concerns regarding HW are primarily focused on spatio-temporal variations of prices for an existing housing stock (e.g., Garbinti, Goupille-Lebret, and Piketty 2021). This approach on capital gains, as prices differences over time, informs on potential financial yields for homeowners. However, it does not pay much attention to the impact

of current socio-spatial dynamics on housing markets as a source of uneven HW accumulation. Indeed, transactions in housing markets induce switches of both people and capital across space, thereby engaging a redistribution of wealth among different groups of homeowners. (Hamnett 2005). Considering that, we elaborate a Price Gap Index (PGI) as the difference between social groups of homeowners in the price gap between purchased and sold properties for each group. The PGI has the advantage of providing a measure of the redistribution of HW.

Our paper brings methodological perspectives in designing a synthetic quantitative framework to describe social stratification in housing markets and allows for the analysis of the impacts of socio-spatial dynamics on the accumulation of HW. This framework allows for the decomposition of HW determinants (on housing characteristics, location, and transaction timing) and balances the effect of redistribution (PGI) with more usual definitions of HW accumulation (i.e., capital gains). The framework is applied to an original dataset of transactions for principal residences, including information on occupational categories (OC) of buyers and sellers. The dataset covers the French region *Bourgogne-Franche-Comté* from 1998 to 2017. This period is relevant for documenting the contrasting HW accumulation between social groups of homeowners. The first decade 1998–2007 is characterized by a substantial increase in the regional housing price index (+130%), while the second decade 2008–2017 witnessed a slight decrease of this index (–3.6%). The quantitative framework is nonetheless general and could also be applied in other regions or time periods.

Starting with a static average price decomposition across OCs, we describe an uneven pattern on the housing

market explaining the social hierarchy of housing prices to understand overall HW inequalities (i). We then test the hypothesis of growing inequalities considering capital gains as an usual suspect of a latent HW accumulation for homeowners benefiting from local housing markets dynamics (ii). By introducing the PGI, we compute the differences in price decomposition between buyers and sellers among OCs (iii). Our results finally demonstrate the importance of socio-spatial upward or downward dynamics in explaining the growing HW inequalities between social groups (iv).

Related literature

We first review the existing literature that uses occupational categories (OCs) as proxies to study social stratification. We then describe how HW accumulation is typically characterized and outline the main patterns of HW across social groups. Finally, we propose the PGI as a new dynamic approach to measure HW accumulation.

Using occupations categories (OCs) for social stratification

Labor-based classifications have a rich historical background in both social and spatial stratification studies (Duncan and Duncan 1955; Blau, Duncan, and Tyree 1978). OCs are often privileged as a proxy of educational attainment, family background, and labor market dynamics, all of which play a role in shaping individuals' incomes (Wright 2000) as well as contributing to various social and cultural group differences (Boterman, Musterd, and Manting 2021).

OCs classification have proven to be more relevant for addressing social stratification than indicators based on activity sectors or income groups (Bourdieu 2005). The stability and versatility of OCs classification make it particularly useful to describe the impact of economic geographies on labor populations in numerous Western countries (Williams 2017). They have been particularly used to capture the transformations of productive systems and the resurgence of polarized labor structures within post-Fordist economies in various case studies such as the United States (?), the EU, and notably France (Vera-Toscano, Fana, and Fernández-Macías 2022).

OCs help to describe the patterns of social inequalities arising from the labor market while and also have a long-standing tradition of documenting the growth of the middle class and the social upgrading processes in global cities (van Ham et al. 2020; Hamnett 2021). OCs have also been crossed with various spatial delineations in the examination of urban segregation dynamics for various cities all over the world (van Ham et al. 2021; Musterd 2020), and more precisely to study gentrification in the French context (Clerval 2022).

A recent literature has highlighted the relevance of linking OCs to housing market outcomes to understand how the latter contribute in wealth inequalities. For instance, authors have tested the hypothesis of whether the housing market creates an independent form of stratification or merely reproduces existing labor market patterns. This was done through national scale analysis (Arundel and Doling 2017; Wind, Lersch, and Dewilde 2017; Smith et al. 2022), or through a spatial analysis of housing markets at a local scale

(Le Goix et al. 2019). Despite these few studies, there is still a lack of research intersecting analyses of the housing market and labor stratification. Our paper aims to contribute in this field by providing empirical evidences and a methodological framework to assess HW accumulation across OCs using housing transaction data.

Capital gains from homeownership

While price differences reveal disparities in Housing wealth (HW), accumulation is usually characterized as capital gains or losses across social groups (?Bach, Calvet, and Sodini 2020). Since it hinges on the appreciation of existing housing stock, it is inherently linked with variations in spatial and temporal conditions within the housing market (Hochstenbach and Arundel 2020).

Variations in housing price over time have led to unequal returns on housing investments across generations (Arundel 2017). On a macro scale, the dynamics of rising prices over three decades have predominantly favored older homeowners. The baby boomer generation, born between 1950 and 1970, has particularly benefited from low purchase prices, favorable housing taxation, and accommodating monetary policies (Forrest and Hirayama 2015; Monnet and Wolf 2017). Conversely, younger generations have faced with high prices and decreasing housing affordability, resulting in negative impacts on housing investment returns (Wind, Lersch, and Dewilde 2017).

Additionally, inflation has been uneven in space. The literature has highlighted the acceleration of metropolization and financialization processes, contributing to higher spatial price disparities. The growing polarization of housing markets, accentuated after the 2008 Global Financial Crisis (GFC), has amplified a long run of concentration of housing wealth in metropolitan areas, benefiting existing homeowners (Gyourko, Mayer, and Sinai 2013). But only few studies have delved into fine-scale analyses, integrating peri-urban and rural spaces, and be able of capturing significant local market effects. Some exceptions has demonstrated that local diverging housing prices occurs through greater social (e.g., Wind and Hedman 2018 for Sweden, Le Goix et al. 2020 for France, and Arundel and Hochstenbach 2020 for the Netherlands), or racial segregation (Fesselmeier, Le, and Seah 2013).

Given that housing is a fixed asset, potential accumulation is also influenced by changing neighborhood conditions and surrounding housing transactions. Arundel and Hochstenbach (2020), suggest that upper classes engage residential strategies to benefit from local accumulation differentials, whereas Hochstenbach and Musterd (2018) emphasize on suburbanization of low-income group in homeownership access in Dutch cities. Similar conclusions are drawn by Le Goix et al. (2019) who examine local price inflation dynamics and social group purchasing behaviors in the suburbs of the Paris region. Wind and Hedman (2018) suggest that upper social groups exploit local price dynamics to accumulate more wealth in their housing pathways.

Translating socio-spatial dynamics into capital gains across social groups isn't always straightforward. Spatial analyses barely models spatial division of housing prices with capital gains across social groups. However, with statistics models, Arundel and Lennartz (2020) emphasize

the relationship between labor social division and housing market position, arguing that housing prices polarization and capital gains opportunities follows the social stratification associated with employment. In an other study, [Arundel and Hochstenbach \(2020\)](#) claim that homeownership access in the most favorable locations for capital gains strongly benefit to upper class and depends on social reproduction through parental wealth transfers.

Specific socio-spatial processes, such as gentrification, can actually amplify capital gain opportunities, even for low socio-economic status. Through an econometric approach on racial groups of buyers in the US, [? claim that Asians buy in areas with higher returns, whereas Black buyers see similar average returns but face with more heterogeneity, influenced by varying local housing price dynamics. Furthermore, long-term inflation contributes to spatial homogenization of housing prices. Consequently, inflation is often more noticeable in low-priced housing and neighborhoods \(Hamnett 2009\). In these instances, the appreciation rate may actually indicate more benefit for low socio-economic status. Nonetheless, in nominal terms, inflation mechanically exacerbates wealth disparities between social groups \(?\)](#).

The effect of transactions on the redistribution of Housing Wealth

In economics, some authors argue that rising prices are roughly neutral in terms of wealth ([Buiter 2010; Bonnet et al. 2015](#)), whereas others contend that latent capital gains increase inequalities between homeowners and renters in terms of both incomes and consumption ([Allègre and Timbeau 2015](#)). In spatial studies, academic literature highlights an increasing spatial divide and polarization of price and capital gains, generally leading to the conclusion that higher socioeconomic status tends to accumulate more HW than other groups. Beyond these debates, most analytical framework tackling the issues of latent capital gains, but they overlook the redistribution of HW in housing markets, that is an aggregate transfer of capital between buyers and sellers during transactions.

There is an established literature in urban economics that examines the role of buyer or seller characteristics in determining housing prices. Recent studies have demonstrated that housing prices systematically vary according to family situation ([Fagereng, Guiso, and Pistaferri 2022](#)), gender ([Waitkus and Minkus 2021](#)), housing intermediary ([Besbris and Korver-Glenn 2022](#)), parental support ([Arundel and Hochstenbach 2020](#)), race ([Bayer et al. 2017](#)), and social group ([Cohen Raviv and Lewin-Epstein 2021](#)). Depending on the case under study and the variables used to control for housing characteristics, these results are alternately interpreted in terms of market segmentation ([Galster 1996](#)), price discrimination ([Ihlanfeldt and Mayock 2009](#)), prejudice, or segregation ([Quillian, Lee, and Honoré 2020](#)). This photography of social stratification through housing prices is however a static approach. We do a step aside by suggesting a dynamic approach that accounts for investment flows by examining the differences across social groups in the existing gap between the average prices of buyers and sellers within each social group.

Regardless of the precise economic behaviors of agents, in an accounting perspective, we consider that redistribution of HW is barely zero-sum across different groups of homeowners because of the realization of a price gap between buyers and sellers. On one hand, if purchasers, on average, move to more expensive housing than the sellers from the same OC, this leads to an overall increase in the HW of the group. On the other hand, if a group buy cheaper housing than they sell, it contributes to an overall decrease in the HW of their OC. Based on this, we assume that differences in the average price gap between buyers and sellers are significant and reflect social divide patterns in HW accumulation that are underestimated by focusing solely on latent capital gains (H1). We also assume that these differences can be partly explained by describing social variations within the spatial hierarchy of housing markets (H2). The Price Gap Index (PGI) has been conceived to address these assumptions with an original decomposition framework adapted from [Gelbach \(2016\)](#).

Methodological framework

This section outlines our general approach to analyze HW across social groups. Building on a classical hedonic model, we present the general principle of [Gelbach \(2016\)](#)'s decomposition, followed by a static decomposition and a dynamic decomposition of HW. The static decomposition describes the positions of homeowners in the housing market for each social groups by examining three main components of prices (*What, Where, When* respectively based on housing characteristics, location, and transaction date). The dynamic decomposition is applied separately for buyers and sellers within each OCs. The differences between these two parts give the PGI. It enables us to understand the impact of redistribution for each OC. Decomposition of PGI offers insight into the spatial upward or downward switch of each OCs in the housing markets explaining differential accumulation of HW. We finally present our approach to estimate capital gains from housing prices variations.

Hedonic models of housing price

For a given housing transaction i of a given period, we note $1[B_i = k]$ the dummy variable equals to one if the buyer is from the OC k and zero otherwise. The equivalent for the seller is noted $1[S_i = k]$. We first consider the following linear model to estimate average housing price differences between the OCs of buyers and sellers:

$$p_i = \tilde{\alpha} + \sum_k \tilde{\phi}_k 1[B_i = k] + \sum_k \tilde{\psi}_k 1[S_i = k] + \tilde{\varepsilon}_i. \quad (1)$$

As usual for OLS regressions, the estimated coefficients $\tilde{\phi}_k$ and $\tilde{\psi}_k$ are respectively the average purchase and selling price differences for each OC k (relative to the reference OC, which we take without loss of generality to be the Retired).

In order to account for housing characteristics, housing locations, and transaction dates, we estimate a more extensive linear model with three additional sets of variables. We note $\mathbf{X}_i$ the set of housing characteristics (e.g., living area, period of construction, number of rooms), η_{c_i} the set of postal code fixed effects, and γ_{t_i} the set of year fixed effects.

This gives:

$$p_i = \alpha + \sum_k \phi_k \mathbf{1}[B_i = k] + \sum_k \psi_k \mathbf{1}[S_i = k] + \mathbf{X}_i' \beta + \eta_{c_i} + \gamma_{t_i} + \varepsilon_i. \quad (2)$$

The vector β represents the marginal hedonic values of the housing characteristics, the introduction of which (together with the space and time fixed effects) changes the value of the coefficients on the OCs, now denoted ϕ_k and ψ_k (i.e. without a tilde). In this more detailed specification, the average differences in purchase and sale prices for each OC are controlled for housing characteristics, location and time difference. The coefficients associated with the buyers' and sellers' OCs (ϕ_k and ψ_k) are now the house price differences not attributable to the other included variables, which we interpret as social segmentation of housing markets.

Gelbach's decomposition of housing prices

From the terminology of Gelbach (2016), the Equation 1 is the *base* model and Equation 2 is the *full* model. The empirical implementation of the decomposition is quite simple. OLS give for each transaction i three partial price predictions depending respectively on housing characteristics ($\mathbf{X}_i' \hat{\beta}$), housing location ($\hat{\eta}_{c_i}$), and transaction year ($\hat{\gamma}_{t_i}$). We separately regress each partial prediction (i) on the dummy variables about the OCs of buyers $\mathbf{1}[B_i = k]$ and (ii) on the dummy variables about the OCs of sellers $\mathbf{1}[S_i = k]$ to obtain six sets of coefficients (two separate regressions for each three partial predictions). We note $\delta_{\phi_k}^{\text{What}}$ the coefficients from the regression of partial predictions from housing characteristics on dummies about the OCs of buyers and $\delta_{\psi_k}^{\text{What}}$ for dummies about the OCs of sellers. The corresponding coefficients are $\delta_{\phi_k}^{\text{Where}}$ and $\delta_{\psi_k}^{\text{Where}}$ for the postal code fixed effects and $\delta_{\phi_k}^{\text{When}}$ and $\delta_{\psi_k}^{\text{When}}$ for year fixed effects.

Gelbach (2016) formally establishes [equation (12), p.522, adapting his notations to ours] that these six terms are equals to the differences between coefficients from Equation 1 and Equation 2, namely:

$$\tilde{\phi}_k - \phi_k = \delta_{\phi_k}^{\text{What}} + \delta_{\phi_k}^{\text{Where}} + \delta_{\phi_k}^{\text{When}} \quad (3)$$

$$\tilde{\psi}_k - \psi_k = \delta_{\psi_k}^{\text{What}} + \delta_{\psi_k}^{\text{Where}} + \delta_{\psi_k}^{\text{When}}. \quad (4)$$

For buyers (3) and sellers (4) of a given OC k , these formulas decompose the differences of coefficients between the base and the full model in terms of housing characteristics (the what component), housing location (the where component), and transaction time (the when component).

Static approach: average price decomposition

A direct application of Gelbach's decomposition provides a picture of the determinants of social differences of HW. The average housing price for buyers and sellers of a given OC k , noted $\bar{p}_k \equiv (\tilde{\phi}_k + \tilde{\psi}_k)/2$, is the average of the three previous decomposition terms plus a residual term:

$$\bar{p}_k = \text{What}_k^S + \text{Where}_k^S + \text{When}_k^S + (\phi_k + \psi_k)/2. \quad (5)$$

This decomposition (5) is the term-by-term average of equations (3) and (4) using the following definitions of: $\text{What}_k^S \equiv (\delta_{\phi_k}^{\text{What}} + \delta_{\psi_k}^{\text{What}})/2$, $\text{Where}_k^S \equiv (\delta_{\phi_k}^{\text{Where}} + \delta_{\psi_k}^{\text{Where}})/2$, and $\text{When}_k^S \equiv (\delta_{\phi_k}^{\text{When}} + \delta_{\psi_k}^{\text{When}})/2$. The last term

at the right side of Equation 5 is the residual part of HW difference, which cannot be attributed to the variables included in the hedonic model and are considered as coming from social segmentation of housing prices. This decomposition is considered as static because the average between buyers and sellers prices is a proxy of the social composition of the owners of housing stock, which do not account for the HW accumulation dynamics that take place during transactions.

Dynamic approach: price gap decomposition

To account for the HW accumulation patterns from housing transactions, we define the Price Gap Index (PGI) that focuses on average price differences between buyers and sellers for each OC (instead of average prices across OCs as before). For a given OC, this amount implies to consider that observing a higher price for buyers than for sellers (ie. a positive PGI) lead to HW accumulation and *vice versa*. From the simple hedonic regression (1), the PGI can be expressed as $PGI_k = \tilde{\phi}_k - \tilde{\psi}_k$ for each OC k .

By subtracting (instead of averaging as in the static approach) equations (3) and (4) of Gelbach's decomposition, we obtain:

$$PGI = \text{What}_k^D + \text{Where}_k^D + \text{When}_k^D + (\phi_k - \psi_k). \quad (6)$$

The decomposition terms are defined from $\text{What}_k^D \equiv \delta_{\phi_k}^{\text{What}} - \delta_{\psi_k}^{\text{What}}$, $\text{Where}_k^D \equiv \delta_{\phi_k}^{\text{Where}} - \delta_{\psi_k}^{\text{Where}}$, and $\text{When}_k^D \equiv \delta_{\phi_k}^{\text{When}} - \delta_{\psi_k}^{\text{When}}$ from the Gelbach's decomposition. The coefficients ϕ_k and ψ_k in the last part of Equation 6 come from the estimation of the full hedonic model (2) with control variables. We interpret this residual part (ie. not explained by the model) at the right side of Equation 6 as the social segmentation of HW accumulation.

Comparison with usual capital gains

Our definition of the PGI complements the more traditional approach to capital gains in HW accumulation. Instead of resulting from the interaction between buyers and sellers of different social groups during transactions, capital gains results from price variations between two transactions for a same social group. The main determinant of capital gains is housing price variations between the purchase and the sale of a house, which can easily integrated in our framework to comparison with PGI. We propose to estimate an auxiliary hedonic model on the same data:

$$p_i = \sum_k (\mu_k + \theta_k \cdot t_i) \times \mathbf{1}[O_i = k] + \epsilon_i. \quad (7)$$

The dummy variable $\mathbf{1}[O_i = k]$ equals one if the buyer *or* the seller is of the OC k and zero otherwise. Hence, the unknown coefficients μ_k and θ_k represent respectively initial housing prices and average annual capital gains in monetary units because the year of the transaction t_i is normalized to be zero at the beginning of the period. As demonstrated in our results, we observe that the piecewise linear specification (7) estimated separately on 1998–2007 and 2008–2017 closely matches the considered data.

Context and data

This section presents the regional context for the application of our decomposition framework and the notarial data about housing transaction we use.

The French region of Bourgogne-Franche-Comté (BFC)

Our application draws on one of the 13 regions of France, covering eight *départements* (see Figure S2). BFC is an intermediate sized region in terms of population and exhibits housing price dynamics quite similar to the French average when excluding *Paris*. The region is characterized by a polycentric network structure of medium and small sized cities, with peri-urban and inland rural areas in between. Suburban areas around major cities such *Dijon*, *Besançon*, *Belfort*, and *Montbéliard*, or close to the Switzerland border, have experienced population growth over the last two decades, as well as job creations in thriving economic sectors (INSEE 2018). In contrast, small towns and rural areas have seen job losses and population decline, in conjunction with an ageing population. About the half of urban areas of the region, especially in the west side, have undergone a shrinking process in recent decades, such that demographic and economic distributions tend to become more polarized.

Compared with other context in western Europe, France has a high ownership rate and the BFC region presents an even higher rate (63.2% instead of 57.1% nationally). This correlates with an old housing stock, a high rate of detached houses (compared to apartments), a high average living surface, and a low average housing price. Nevertheless, the dynamics of homeownership are close to the national ones. Whereas most of spatially explicit works on housing markets have focused on attractive metropolitan areas, more common and less attractive places, encompassing urban, peri-urban, and rural settings are under-studied. Finally, the two decades under study (1998-2007 and 2008-2017) allow us to investigate the differentiated patterns of HW distribution implied by the price trend break (increasing then slightly decreasing) of the 2008 financial crisis.

Notarial data about housing transactions

In France, notaries are responsible for registering property rights, and our analysis relies on their records of housing transactions for the period 1998–2017. These data distinguish main residences, and the OC of the homeowner according to the half century-old classification of *groupes socio-économiques* (Desrosières and Thévenot 1988). These OCs come from the French official classification of jobs, grouped in six aggregated items, plus two items (Inactive and Retired). This classification is processed by hand by notaries from an open question about the main job in the sale deeds. We groups them in eight items: Farmer, Independent, Senior Staff, Intermediate, Clerk, Blue Collar, Inactive, and Retired. The categories of Farmer, Clerk, and Blue Collar are well known in the international literature about social stratification. Independent includes business owners in addition to the more usual craftsmen and merchants, Senior Staff includes executives, managers, and other intellectual professions, and Intermediate is more defined by elimination from other OCs and may contain diverse back-office jobs, foremen, freelance nurses, self-employed sports instructors, or technicians.

After usual data cleaning and selection operations,* we obtain a final sample of 295,199 housing transactions. The bottom panel of Table S1 displays descriptive statistics about

average housing price differences and the shares of each OC across the buyers and sellers. Except for Retired with twice as many sellers, all other OCs have a greater rate of buyers. These transaction data deliver full information on housing characteristics, time, and location, which are used as covariates to control for housing prices in order to describe the three dimensions of housings (the *What*, *When*, and *Where* components of the decomposition).

Results

In this section, static social hierarchy of HW according to the decomposition of housing prices are describe. Next, we report capital gains estimated from auxiliary housing price regressions (Equation 7) and compare them with the dynamic PGI decomposition. Finally, we discuss the socio-spatial patterns of HW accumulation according to the PGI.

Social hierarchy of Housing Wealth

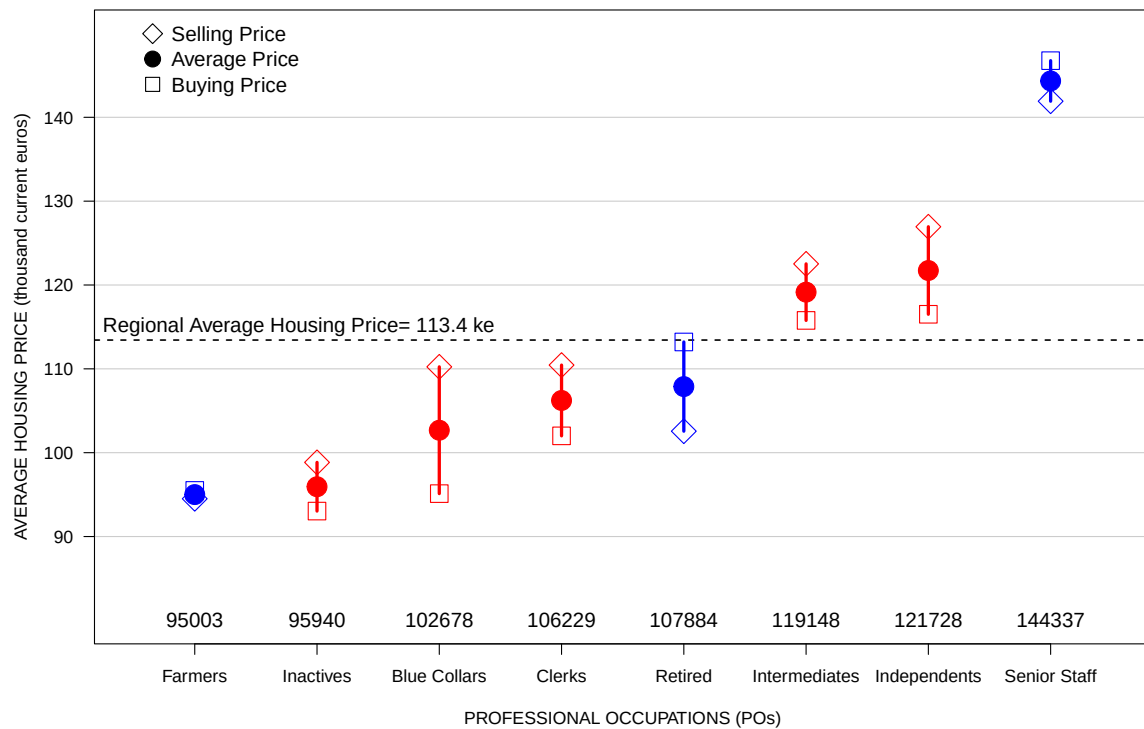
From a static perspective, we examine the social hierarchy of HW through the average housing price differences observed among social groups. Figure 1 shows in filled dots the average housing price $\bar{p}_k$ from Equation 1 without any control variables. As expected, the hierarchy observed on the housing market mirrors the one of the labor market. OCs are ordered from Farmer to Senior Staff (with average difference of 50k euros by housing unit), with Clerks and Retired that hold an intermediate position. This hierarchy is stable between the two decades (Figure S4), which justifies that we do not differentiate them from a static perspective.

Table 1 reports the static decomposition of housing price differences from regional averages across OCs (Equation 2). It shows the *What*, *Where*, *When* components, and an unexplained part from the remaining OC dummies (i.e., the social segmentation of the housing market). The raw regression coefficients are reported in Table S2, spatial and time fixed effects obtained from OLS models are reported in Figures S3 and S4. The *What* variables[†] are all significant. Including these terms jointly with spatial (*Where*) and temporal (*When*) fixed effects produces a R^2 of about 73%, and decreases the absolute values associated with the OCs.

The hierarchy of the different components varies across OCs (Table 1). HW of S. Staff is higher than other OCs, mostly because of better location (35%) and better housing characteristics (25%). At the opposite, the location part is lower than the average for Farmer, B. Collars, Inactive, and Retired, suggesting that these OCs tend to be involved (both at buyers and sellers) in transactions away from best prices locations (Figure S4). Clerk does not have a significant *Where* component, which indicates that they are better located than the other low-status OCs. However, jointly with Retired, Inactive, and Intermediates, their intrinsic housing characteristics are of lower quality than the regional average (whereas Farmer and B. Collar have better characteristics).

*We keep only transactions of principal residences between private persons, and drop transactions with omitted or extreme values on the variables of interest, i.e., outside their 2.5% interquartile ranges.

[†]Housing type, construction date, living area, lot size, number of pieces, and the presence of dependencies (e.g., cave, parking, balcony)

Figure 1. Average housing prices and housing price gaps between OCs

Notes: This figure displays the average housing prices $\bar{p}_k$ by filled dots (reported in euros at the bottom). They are computed as the average of buying (empty square) and selling (empty diamond) prices, respectively noted $\tilde{p}$ and $\hat{p}$ in Equation 1. The dots and lines are colored in red when the selling price is higher than the buyer price (i.e., a negative PGI) and in blue otherwise. All the reported values are significantly different from zero at 95% confidence. Sources: Perval (Notaires).

Lastly, despite the inclusion of many explanatory variables, time, and space fixed effects, the OCs dummies still have a strong statistical significance in the extensive hedonic model (4) with high unexplained components (in absolute values) for each OC. They reach respectively +13k euros (41.9% of average price) for S. Staff and −6.25k euros (58.2%) for B. Collar. This indicates *ceteris paribus* that housing markets are socially segmented according to the OCs of homeowners.

Capital gains: an insufficient measure to capture inequalities

According to the scientific literature, the main driver of HW accumulation inequalities is attributed to latent capital gains realized by homeowners when housing price fluctuate. In these frameworks, capital gains represent a snapshot of prices, year after year, of an existing housing stock.

To compare our PGI approach in the next section with usual capital gains, we estimate initial housing prices and their average variations over time for each OC and each decade (Equation 7). Regressions of housing prices on OC fixed effects and their interaction with time trends provide the piecewise linear trends reported in Figure S1. These trends reveal the housing price break following the 2008 Global Financial Crisis (GFC), with variations allowed between OCs. Table S3 reports the raw coefficients from each auxiliary regression, with R^2 values ranging from 0.23 for the first decade to 0.09 for the second. It appears that the piecewise linear time trends exhibit a striking

symmetry across OCs.[‡] The results in Table S3 are robust to alternative specifications. Including postal code fixed effects does not change the trends, indicating robustness to spatial heterogeneity at the postal code level.

From the intercepts and slopes of the time trends in Figure S1, Table S2 displays average housing prices, annual nominal gains, and annual growth rates for each OC and each decade. Average housing prices follow the socio-economic status hierarchy, ranging from 54.53k euros for B. Collar to 104k euros for S. Staff in 1998. The housing price boom of the first decade has enlarged the distribution of average housing prices, as evidenced by differences in average nominal gains. These gains are significantly higher than the regional average for Independent and S. Staff (of respectively +920 and +540 euros by year), and lower for other OCs (only Farmer and Inactive do not have significant differences with the regional average). In contrast, when capital gains are expressed as growth rates (% of initial housing prices), the hierarchy among social group shifts: Senior Staff earns 10.42% each year whereas Blue Collar earns 17.13%. In other words, HW accumulation across OCs is regressive in nominal terms but progressive in relative terms, although the latter is less statistically significant.

The second row panel of Table S2 shows that housing price differences across OCs are somewhat more marked during the second decade. The differences with the regional

[‡]The higher variability around the trend of Farmer is due to the small quantity of housing transactions.

Table 1. Decomposition of average housing prices across OCs

	<i>Differences across Occupational Categories (OCs)</i>							
	Farmer	Indep.	S.Staff	Interm.	Clerk	B.Collar	Inactive	Retired
Housing Price (Region= 113.4)	-18.43*** (-16.2%)	+8.29*** (7.3%)	+30.9*** (27.2%)	+5.71*** (5%)	-7.2*** (-6.4%)	-10.76*** (-9.5%)	-17.49*** (-15.4%)	-5.55*** (-4.9%)
What _k ^A	+1.71 [-9.3%]	+4.8*** [57.9%]	+7.49*** [24.2%]	-2.19*** [-38.3%]	-3.96*** [55%]	+2.49*** [-23.1%]	-5.32*** [30.4%]	-2.86*** [51.5%]
When _k ^A	-2.18*** [11.8%]	-0.13 [-1.6%]	-0.24** [-0.8%]	+1.01*** [17.7%]	+0.34*** [-4.8%]	+0.09 [-0.8%]	-3.76*** [21.5%]	-0.2** [3.6%]
Where _k ^A	-11.37*** [61.7%]	+0.49** [5.9%]	+10.71*** [34.7%]	+5.53*** [96.8%]	-0.23 [3.2%]	-7.08*** [65.8%]	-3.92*** [22.4%]	-3.93*** [70.8%]
$(\phi_k + \psi_k)/2$	-6.6*** [35.8%]	+3.14*** [37.8%]	+12.95*** [41.9%]	+1.36*** [23.8%]	-3.36*** [46.6%]	-6.26*** [58.2%]	-4.49*** [25.6%]	+1.44*** [-25.9%]

*** denotes different from zero with at least 99% confidence, ** for 95% and * for 90%.

Notes: The top panel of the table reports the average housing prices differences across OCs (expressed in k euros of differences from the regional average). The bottom panel reports the Gelbach (2016)'s decomposition of these differences across OCs, both expressed in thousand euros and in percent of the first row. The respective components concern housing characteristic (What_k^A), transaction year (When_k^A), housing location (Where_k^A). The residual OC effect noted $(\phi_k + \psi_k)/2$ corresponds to the unexplained part of differences (Table S1 reports the formulas we use).

Source: Perval (Notaires).

average range from -11.37k euros for B. Collar to +34.68k euros for Senior Staff, covering more than one-third of the average initial price. While the average nominal gains and growth rates are slightly negative at the regional scale (albeit not significant), three OCs—Farmer, Independent, and B. Collar—show significantly different nominal gains. Independents also exhibit a significantly higher growth rate, at 1.29% per year. Surprisingly, nominal gains and growth rates become more homogeneous across OCs after 2008. In short, price boom increase the dispersion in nominal terms, while the stagnation post-2008 has modestly reduced it.

Price Gap Index: a dynamic approach of Housing Wealth accumulation

While some academic literature highlights worsening inequalities in HW due to housing price variations, our measure of capital gains does not fully support this conclusion. We found that disparities in capital gains between social groups were less regressive than those observed in static HW stratification approaches. Unexpectedly, there was a catching-up effect in ratios and no significant reinforcement of capital gains inequalities following the 2008 GFC. To accurately capture the growing disparities in HW between social groups, it's essential to consider not only the appreciation of the housing stock but also the redistribution of HW that occurs during transactions. This redistribution is notably influenced by varying social housing pathways. Our dynamic approach, through the Price Gap Index (PGI), captures this redistribution.

PGI does not simply follow the uniform trends of capital gains but reveals a more regressive pattern between OCs. Back to Figure 1, the buying and selling prices used to compute the PGI for each OC k (ie. $\tilde{\phi}_k - \tilde{\psi}_k$) are statistically significant (see also top panel of Table 2). OCs are distinguished between those in red that sell housings at a higher price than they buy (Inactive, B. Collar, Clerk, Intermediate, and Independent), and those in blue that buy at a higher price than they sell (Retired and S. Staff). Intermediate and low social groups have a negative PGI, which mean than in average they display a

HW dis-accumulation. Indep., Interm., Clerk, B.Collar, and Inactive decrease their HW by respectively 10, 7, 8, 15, and 6k euro by transaction according to Table 2. At the opposite S. Staff and Retired have a positive PGI, their HW increases by respectively 5 and 11k euro per transaction. Unexplained parts of PGI accounts for about one-third of price differences within OCs (in absolute values), which are positive for Retired, S. Staff, Farmer, and Inactive. Once again we interpret these as a socail segmentation effects in the housing market, which produce *ceteris paribus* HW losses for the others OCs (Independent, Intermediate, Clerks, and B. Collar).

Our framework enables the decomposition of the PGI and elucidates social trends in HW accumulation. The *What* component of housing price differences within OCs reflects changes in the intrinsic quality of housing. It is positive for buyers who upgrade the quality of their housing compared to sellers (e.g., Farmer, Independent, Senior Staff, and Intermediate), and negative for others (see Table 2). The *Where* component captures the effect of location: a positive value indicates an upward shift in location quality, while a negative value signifies a decline. This upward gap in the social hierarchy of HW is only observed for Retired, altering their position in comparison with static social hierarchy of HW, whereas all other OCs experience a downward shift in location, especially for low economic status. The *When* component gives the effect of transaction timing (differences between years of purchase and sale) and is generally less influential than the two other components.

Socio-spatial patterns of Housing Wealth accumulation

PGI provides valuable insights to link social hierarchy with spatial dynamics of OCs. Delving into the socio-spatial patterns of HW accumulation, Table 3 explores the correlation between the number of housing transactions and the differences between the numbers buyers and sellers with the spatial fixed effects of the hedonic housing price equation (η_{ci} of Equation 2). Static and dynamic approaches

Table 2. Decomposition of average housing price dynamics across OCs

	Differences across Occupational Categories (OCs)							
	Farmer	Indep.	S.Staff	Interm.	Clerk	B.Collar	Inactive	Retired
Price Diff.	+0.99*** (0.9%)	−10.44*** (−9.2%)	+4.84*** (4.3%)	−6.74*** (−5.9%)	−8.44*** (−7.4%)	−15.12*** (−13.3%)	−5.81*** (−5.1%)	+10.64*** (9.4%)
What _k ^P	+1.92*** [193.4%]	+1.89*** [−18.1%]	+5.89*** [121.7%]	+1.35*** [−20%]	−2.59*** [30.7%]	−3.01*** [19.9%]	−10.49*** [180.7%]	−4.09*** [−38.5%]
When _k ^P	−0.29*** [−29%]	−0.92*** [8.8%]	+0.4*** [8.3%]	−0.57*** [8.5%]	−0.18*** [2.1%]	−0.97*** [6.4%]	+2.6*** [−44.8%]	−0.26*** [−2.5%]
Where _k ^P	−3.33*** [−335.3%]	−5.4*** [51.7%]	−2.83*** [−58.4%]	−2.82*** [41.9%]	−2.25*** [26.7%]	−5.35*** [35.4%]	+1.34*** [−23%]	+3.65*** [34.3%]
($\phi_k^P - \psi_k^P$)	+2.69*** [270.9%]	−6.02*** [57.6%]	+1.38*** [28.4%]	−4.69*** [69.6%]	−3.42*** [40.5%]	−5.79*** [38.3%]	+0.74*** [−12.8%]	+11.35*** [106.7%]

*** denotes different from zero with at least 99% confidence, ** for 95% and * for 90%.
Notes: The top panel reports average price difference between buying and selling prices (in k euros) and the bottom panel reports the Gelbach (2016)’s decomposition both in k euros and in percent of the first row. The components are about housing characteristic (What_k^P), transaction year (When_k^P), housing location (Where_k^P), and direct OC effect ($\phi_k^P - \psi_k^P$) (i.e., the unexplained part).
Source: Perval (Notaires).

of housing transactions show contrasted spatial patterns of HW.

For instance, the average number of transactions involving Retired buyers or sellers tends to be higher in less expensive spatial units, while the number of Retired buyers exceeds the number of Retired in more expensive postal codes. This observation highlights the fundamental spatial switch in the HW of this group towards more central urban areas.

In contrast, B. Collars, Clerks and Farmers are located and tend to move in less expensive spatial units with an average dis-accumulation of respectively, −15.12k, −7.2k, and −18.43k euros relatively to the regional average (Table 2). The dynamic approach reveals a more pronounced negative correlation for B. Collar and Clerks compared to the static approach (whereas the reverse is true for Farmers). This underscores the complementary of both approaches to illustrate trends of displacement towards more peripheral areas, aligning with findings in metropolitan contexts (Hochstenbach and Musterd 2018).

S. Staff, on the other hand, are over-represented in more expensive areas, both in static and dynamic approaches, though the correlations are stronger in the former approach. This spatial diversification can be interpreted in two ways. First, it may reflect current gentrification in urban areas where social competition with wealthier retired households drives them to more affordable and social mixed neighborhoods. Second, particularly for new entrants to the housing market, as other active OCs, there is also a notable trend of peripheralization.

The GFC of 2008 does not alter the signs of the correlation coefficients of the static approach, but it does shift the sign from negative to positive in the dynamic approach for Interm. and S. Staff that were impacted by the crisis in terms of location. The bottom panel of Table 3 reveals that GFC intensified the correlation between the static and dynamic location patterns of HW, where OCs can be divided into two main categories: those who move to spatial units where they are less numerous–Retired and Independents– and all other OCs, who tend to either remain in or move to areas that reflect their established presence. For the latter, it is logically very strong for farmers, but we also observe a trend towards spatial homogenization for Clerks, Inactive, and B. Collars,

while Intermediates and S. Staff have a spatial footprint that is diversifying.

Discussion and conclusion

A major contribution of this paper is development of a quantitative approach to catch the effects of HW redistribution on social inequalities. Furthermore, our decomposition framework enables an understanding of the influence of socio-spatial changes in the worsening of HW inequalities. To achieve this, we have based our analysis on long-standing social groups stratified in the labor market. While OCs are merely control variables in most studies on HW inequalities, our conceptual framework put them at the forefront of the analysis and provides a complementary picture of HW redistribution operated by the housing markets. Instead of looking for a market price of a given housing, our focus shifts towards understanding housing price-induced inequalities. This approach could benefit to be replicated for other regions, social classifications, or time periods.

Our analysis on capital gains reach the conclusion of Burbidge (2000) in his study on Sydney metro area. Prolonged trends of price inflation increase mechanically HW inequalities across OCs on nominal terms. However, as it has been observed in the literature (Le Goix et al. 2019), inflation is proportionally more intense in the lower-priced segments of the market, fostering a relative homogenization of HW accumulation in percentage over recent decades.

This paper introduces a price gap index (PGI) providing new insights to study HW accumulation with housing transactions. For a given social group a positive PGI indicates a *ceteris paribus* increase of HW, and a negative PGI indicates a decrease. The impactf of PGI on both social and generational inequalities in HW accumulation is significant and aligns with findings in literature. Statistical results confirms the importance of taking into account this price difference in order to better measure the effects of market dynamics on HW. Ultimately, we observe an increase in inequalities, with the social structure of labor and the housing market generally exhibiting a comparable hierarchy.

Table 3. Correlations between the spatial distribution of housing prices and the location of housing transactions

	<i>Occupational Categories (OCs)</i>							
	Farmers	Indep.	S.Staff	Interm.	Clerks	B.Collar	Inactive	Retired
Correlation btw number of transaction and the postal code fixed effects								
Before 2008 crisis	-0.162	0.074	0.437	0.355	-0.062	-0.096	-0.293	-0.389
After 2008 crisis	-0.155	0.065	0.391	0.451	-0.131	-0.037	-0.226	-0.43
Correlation btw number of buyers minus sellers and the postal code fixed effects								
Before 2008 crisis	-0.052	-0.104	-0.014	-0.04	-0.118	-0.065	0.049	0.267
After 2008 crisis	-0.058	-0.038	0.058	0.062	-0.208	-0.155	-0.115	0.325
Correlation btw number of transaction and number of buyer minus sellers								
Before 2008 crisis	0.15	-0.242	0.392	0.195	0.057	0.037	-0.008	-0.457
After 2008 crisis	0.559	-0.308	0.045	0.116	0.272	0.154	0.717	-0.59

Notes: This Table displays the correlations between the number of housing transactions from a static (average of the number of buyers and sellers for each OC and each time period) and a dynamic (differences between the numbers of buyers and sellers for each OC and each time period) perspective and the Where^S component of the decomposition. The two first rows show the social position in the spatial hierarchy of the housing market before and after 2008, the closer the correlation approaches one, the more the OC's homeowners are located in expensive areas. The two following rows represent the dynamic approach before and after 2008. The closer the correlation approaches one, the more the OC's buyers move in more expensive areas than OC's sellers. The two last rows report the correlation between static and dynamic numbers of housing transactions.

Source: Perval (Notaires).

Our study is mostly descriptive and does not attempt to disentangle the causes of PGI differentiation. We only refer to the influence of the socio-economic status of buyers and sellers on the housing market. With our framework we shown that PGI is significantly explained through the variations between buyer and sellers in location within the spatial hierarchy of housing markets. Nevertheless, considering social groups give insights, it does not necessarily lead to a clear interpretation of the mechanism. Our results should alongside other studies to uncover heuristic phenomenon. PGI would definitely have several meaning in terms of capital circulation, investments strategies, and consumption across social groups, but also in terms of housing markets segmentation. This open several hypothesis, including capital switching between housing and other markets as well as between places in residential pathway of social groups. This paved the way for future research looking for a better understanding of the inequalities captured by PGI.

Other limitations include the focus on this specific French region and the data constraints raises the usual concern regarding the ability to generalize the findings to broader observations on inequalities. The choice of BFC enables to characterize ordinary instances of inequality in territories comprised of small and medium-sized urban networks. This case study enriches the existing body of literature that predominantly concentrates on metropolitan regions. This approach is pivotal as many Western countries have such under-studied mixed areas. It would also be worthwhile to extend the analysis by including more up-to-date data.

Lastly, the present paper is an initial step for further application. Ongoing work aims to measure wealth inequalities at various spatial scales across France by integrating these components into a more complete measure of a net HW accumulation, including the effects of local variations in homeowners population. For instance, capital gains will be extrapolated to a population that remains in same location and PGI will be related with broader residential mobilities, allowing us to extrapolate our results

to the stock of local newcomers (buyer prices) and outgoing homeowners (sellers prices).

References

- Adkins, Lisa, Melinda Cooper, and Martijn Konings. 2019. "Class in the 21st century: Asset inflation and the new logic of inequality." *Environment and Planning A: Economy and Space* 548–572.
- Allègre, Guillaume, and Xavier Timbeau. 2015. "Le capital-logement ne contribue-t-il vraiment pas aux inégalités ? Réponse à O. Bonnet, P.-H. Bono, G. Chapelle et É. Wasmer (2014)." *Revue de l'OFCE* 137 (1): 83–95. Place: Paris Publisher: OFCE.
- Arundel, Rowan. 2017. "Equity inequity: Housing wealth inequality, inter and intra-generational divergences, and the rise of private landlordism." *Housing, Theory and Society* 34 (2): 176–200.
- Arundel, Rowan, and John Doling. 2017. "The end of mass homeownership? Changes in labour markets and housing tenure opportunities across Europe." *Journal of Housing and the Built Environment* 32 (4): 649–672.
- Arundel, Rowan, and Cody Hochstenbach. 2020. "Divided access and the spatial polarization of housing wealth." *Urban Geography* 41 (4): 497–523.
- Arundel, Rowan, and Christian Lennartz. 2020. "Housing market dualization: linking insider–outsider divides in employment and housing outcomes." *Housing Studies* 35 (8): 1390–1414.
- Bach, Laurent, Laurent E Calvet, and Paolo Sodini. 2020. "Rich pickings? Risk, return, and skill in household wealth." *American Economic Review* 110 (9): 2703–47.
- Badcock, B A. 1989. "Homeownership and the Accumulation of Real Wealth." *Environment and Planning D: Society and Space* 7 (1): 69–91.
- Bayer, Patrick, Marcus Casey, Fernando Ferreira, and Robert McMillan. 2017. "Racial and ethnic price differentials in the housing market." *Journal of Urban Economics* 102: 91–105.
- Besbris, Max, and Elizabeth Korver-Glenn. 2022. "Value fluidity and value anchoring: race, intermediaries and valuation in two

- housing markets." *Socio-Economic Review*.
- Blau, P.M., O.D. Duncan, and A. Tyree. 1978. *American Occupational Structure*. Free Press.
- Bonnet, Odran, Pierre-Henri Bono, Guillaume Chapelle, and Etienne Wasmer. 2015. "Réflexions sur le logement, la hausse des prix de l'immobilier et les inégalités en réponse à l'ouvrage de Thomas Piketty, *Le capital au XXI^e siècle*." *Revue d'Économie Politique* 125: 317–346.
- Boterman, Willem R., Sako Musterd, and Dorien Manting. 2021. "Multiple dimensions of residential segregation. The case of the metropolitan area of Amsterdam." *Urban Geography* 42 (4): 481–506.
- Bourdieu, Pierre. 2005. *The social structures of the economy*. Polity.
- Buiter, Willem H. 2010. "Housing Wealth Isn't Wealth." *Economics: The Open-Access, Open-Assessment E-Journal* 4.
- Burbidge, Andrew. 2000. "Capital Gains, Homeownership and Economic Inequality." 23.
- Clerval, Anne. 2022. "Gentrification and social classes in Paris, 1982–2008." *Urban Geography* 43 (1): 34–58.
- Cohen Raviv, Or, and Noah Lewin-Epstein. 2021. "Homeownership regimes and class inequality among young adults." *International Journal of Comparative Sociology* 62 (5): 404–434.
- Desrosières, Alain, and Laurent Thévenot. 1988. *Les catégories socioprofessionnelles*. Nachdr. ed. Berkeley: Coll Repères, Ed. La Découverte, Paris.
- Duncan, Otis Dudley, and Beverly Duncan. 1955. "Residential Distribution and Occupational Stratification." *American Journal of Sociology* 60 (5): 493–503.
- Fagereng, Andreas, Luigi Guiso, and Luigi Pistaferri. 2022. *Assortative Mating and Wealth Inequality*. Technical Report. National Bureau of Economic Research.
- Fernandez, Rodrigo, and Manuel B. Aalbers. 2017. "Housing and Capital in the Twenty-first Century: Realigning Housing Studies and Political Economy." *Housing, Theory and Society* 34 (2): 151–158.
- Fesselmeyer, Eric, Kien T. Le, and Kiat Ying Seah. 2013. "Changes in the white–black house value distribution gap from 1997 to 2005." *Regional Science and Urban Economics* 43 (1): 132–141.
- Forrest, Ray, and Yosuke Hirayama. 2015. "The financialisation of the social project: Embedded liberalism, neoliberalism and home ownership." *Urban Studies* 52 (2): 233–244.
- Galster, George. 1996. "William Grigsby and the analysis of housing sub-markets and filtering." *Urban Studies* 33 (10): 1797–1805.
- Garbinti, Bertrand, Jonathan Goupille-Lebret, and Thomas Piketty. 2021. "Accounting for wealth-inequality dynamics: methods, estimates, and simulations for France." *Journal of the European Economic Association* 19 (1): 620–663.
- Gelbach, Jonah B. 2016. "When do covariates matter? And which ones, and how much?" *Journal of Labor Economics* 34 (2): 509–543.
- Gyourko, Joseph, Christopher Mayer, and Todd Sinai. 2013. "Superstar Cities." *American Economic Journal: Economic Policy* 5 (4): 167–199.
- Hamnett, Chris. 2005. *Winners and Losers*. Routledge. OCLC: 1191059827.
- Hamnett, Chris. 2009. "Spatially Displaced Demand and the Changing Geography of House Prices in London, 1995–2006." *Housing Studies* 24 (3): 301–320.
- Hamnett, Chris. 2021. "Veni, vidi, gentri? – Social class change in London and Paris: gentrification cause or consequence?" *Urban Geography* 42 (8): 1045–1053.
- Hochstenbach, Cody, and Rowan Arundel. 2020. "Spatial housing market polarisation: National and urban dynamics of diverging house values." *Transactions of the Institute of British Geographers* 45 (2): 464–482.
- Hochstenbach, Cody, and Sako Musterd. 2018. "Gentrification and the suburbanization of poverty: changing urban geographies through boom and bust periods." *Urban Geography* 39 (1): 26–53.
- Ihlanfeldt, Keith, and Tom Mayock. 2009. "Price discrimination in the housing market." *Journal of Urban Economics* 66 (2): 125–140.
- INSEE. 2018. "Déclin marqué ou affirmation de leur rôle central, les destins contrastés des petites villes de Bourgogne-Franche-Comté." *Insee Analyses* (31): 4.
- Le Goix, Renaud, Laure Casanova Enault, Loïc Bonneval, Thibault Le Corre, Eliza Benites-Gambirazio, Guilhem Boulay, William Kutz, Natacha Aveline-Dubach, Julien Migozzi, and Ronan Ysebaert. 2020. "Housing (In)Equity and the Spatial Dynamics of Homeownership in France: A Research Agenda." *Tijdschrift voor economische en sociale geografie* 124: 60.
- Le Goix, Renaud, Timothée Giraud, Robin Cura, Thibault Le Corre, and Julien Migozzi. 2019. "Who sells to whom in the suburbs? Home price inflation and the dynamics of sellers and buyers in the metropolitan region of Paris, 1996–2012." *PLoS One, Public Library of Science* 14 (3): 42.
- Monnet, Eric, and Clara Wolf. 2017. "Demographic cycles, migration and housing investment." *Journal of Housing Economics* 38: 38–49.
- Musterd, Sako, ed. 2020. *Handbook of urban segregation*. Research handbooks in urban studies. Cheltenham, UK Northampton, MA, USA: Edward Elgar Publishing.
- Piketty, Thomas, and Gabriel Zucman. 2014. "Capital is Back: Wealth-Income Ratios in Rich Countries 1700–2010*." *The Quarterly Journal of Economics* 129 (3): 1255–1310.
- Quillian, Lincoln, John J Lee, and Brandon Honoré. 2020. "Racial discrimination in the US housing and mortgage lending markets: a quantitative review of trends, 1976–2016." *Race and Social Problems* 12 (1): 13–28.
- Renaud, Bertrand, and Kyung-Hwan Kim. 2008. "The Global Housing Price Boom and its Aftermath." 18.
- Rex, John, and Robert Moore. 1967. *Race, Community and Conflict: A Study of Sparkbrook*. Oxford university press ed. London ; New York.
- Ryan-Collins, Josh, and Cameron Murray. 2021. "When homes earn more than jobs: the rentierization of the Australian housing market." *Housing Studies* 1–30.
- Saunders, Peter. 1984. "Beyond housing classes: the sociological significance of private property rights in means of consumption." *International Journal of Urban and Regional Research* 8 (2): 202–227.
- Schwartz, Herman, and Leonard Seabrooke. 2008. "Varieties of residential capitalism in the international political economy : old welfare states and the new politics of housing." *Comparative European Politics* 6 (3): 237–261.

- Smith, Susan J., William A. V. Clark, Rachel Ong ViforJ, Gavin A. Wood, William Lisowski, and N. T. Khuong Truong. 2022. "Housing and economic inequality in the long run: The retreat of owner occupation." *Economy and Society* 51 (2): 161–186.
- Thorns, D C. 1989. "The Impact of Homeownership and Capital Gains upon Class and Consumption Sectors." *Environment and Planning D: Society and Space* 7 (3): 293–312.
- van Ham, Maarten, Tiit Tammaru, Rūta Ubarevičienė, and Heleen Janssen, eds. 2021. *Urban Socio-Economic Segregation and Income Inequality: A Global Perspective*. The Urban Book Series. Cham: Springer International Publishing.
- van Ham, Maarten, Masaya Uesugi, Tiit Tammaru, David Manley, and Heleen Janssen. 2020. "Changing occupational structures and residential segregation in New York, London and Tokyo." *Nature Human Behaviour* 4 (11): 1124–1134.
- Vera-Toscano, Esperanza, Marta Fana, and Enrique Fernández-Macías. 2022. *Regional heterogeneity in occupational change: Using Census data to investigate employment polarisation and upgrading at NUTS-3 level*. Technical Report. JRC Working Papers Series on Labour, Education and Technology.
- Waitkus, Nora, and Lara Minkus. 2021. "Investigating the gender wealth gap across occupational classes." *Feminist Economics* 27 (4): 114–147.
- Williams, Mark. 2017. "Occupational Stratification in Contemporary Britain: Occupational Class and the Wage Structure in the Wake of the Great Recession." *Sociology* 51 (6): 1299–1317.
- Wind, Barend, and Lina Hedman. 2018. "The uneven distribution of capital gains in times of socio-spatial inequality: Evidence from Swedish housing pathways between 1995 and 2010." *Urban Studies* 55 (12): 2721–2742.
- Wind, Barend, Philipp Lersch, and Caroline Dewilde. 2017. "The distribution of housing wealth in 16 European countries: accounting for institutional differences." *Journal of Housing and the Built Environment* 32 (4): 625–647.
- Wright, Erik Olin. 2000. *Class Counts*. Studies in Marxism and Social Theory. Cambridge University Press.